

Anurag Pradhan

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#)

SUMMARY

Prefinal-year Computer Science student at VIT Chennai with hands-on experience in Deep Learning and Machine Learning. Skilled in designing, training, and deploying efficient CNNs, Transformers, LLMs and Vision-Language Models, with expertise in PyTorch, data preprocessing, optimization, and on-device inference. Published paper at IJCNLP-AACL 2025, focusing on accessibility-driven evaluation of SmolVLM2-500M-Video-Instruct.

EDUCATION

- Vellore Institute of Technology** Chennai, India
Bachelor of Technology in Computer Science; CGPA: 7.83/10 *August 2023 – June 2027*
- Sri Venkateswara Jr. College** Visakhapatnam, India
Class 12th (MPC); Score: 88.4/100 *July 2021 – June 2023*
- Sri Chaitanya Techno School** Visakhapatnam, India
Class 10th Score: 93.6/100 *February 2020 – March 2021*

SKILLS SUMMARY

- Programming Languages:** Python, C, C++, Java
- Core Expertise:** Deep Learning, Computer Vision, CNNs, Vision Transformers, Image Processing, Sequence Models (LSTM/GRU/RNN), Generative AI, Large Language Models
- Frameworks & Libraries:** PyTorch, Hugging Face Transformers, LLaMA, BEiT, BLIP, Llama.cpp
- Tools & Platforms:** Git, Docker, CUDA, OpenCV, ONNX, LangChain, Hugging Face Hub, Weights and Biases
- Soft Skills:** Leadership, Technical Writing, Event Management, Time Management

EXPERIENCE

- National Aluminium Company Ltd (NALCO)** Damanjodi, Odisha
Computer Science Intern *June 2025 – July 2025*
 - Contribution:** Worked on RespiGuard AI, an AI-based respiratory safety system for mining environments. Improved hazard detection reliability using deep-learning-based visual analysis.
 - Tech Stack:** Custom YOLOv5, PyTorch, Data Augmentation, Model Evaluation
- Team Ignition, VIT Chennai** On Site
Flight Software Lead *Apr 2024 – Present*
 - Contribution:** Developed flight software for model rockets, including fin actuator control, stability algorithms, onboard video stabilization, and Kalman-based state estimation.
 - Tech Stack:** Arduino, C/C++, PID Control, Image Processing, Deep Learning

PROJECTS

- DrishtiAI — Android App:** Developed a multimodal Android application that performs on-device image and video question answering by integrating SmolVLM2-500M-Video-Instruct via llama.cpp. Implemented custom video frame extraction and RGB processing pipelines to enable real-time, offline visual content analysis.
Tech: VLMs, Quantization, Llama.cpp, GGUF, Kotlin
- SentinelVision — Real-time Action Detection System:** Developed a multi-modal deep learning system for detecting critical actions (fight, collapse, normal) in surveillance videos with 95 percent accuracy. Leveraged bidirectional causal temporal attention for video reasoning, VQ-VAE-based optical flow tokenization (1000× compression), and multi-task learning across classification, bounding box regression, and temporal localization with uncertainty quantification [[GitHub Repo](#)]
Tech: PyTorch, Temporal Feature Extraction, Transformers, VQ-VAE, Optical Flow, MediaPipe, Causal Attention, Uncertainty Quantification
- Brain Tumor Segmentation:** Built a BEiT-transformer segmentation pipeline with skip connections and attention-driven decoding. Trained with Albumentations and CE-Dice hybrid loss for stable convergence. [[GitHub Repo](#)]
Tech: Python, PyTorch, Vision Transformers, BEiT, Semantic Segmentation
- JanaSathi: Odia E-Governance Chatbot:** Designed a bilingual RAG-based chatbot for Odisha government schemes using OpenRouter LLMs and MiniLM embeddings. Integrated Odia translation with Cohere Command-R+. Achieved 2nd place in AMD-sponsored OdiaGenAI Hackathon 2025. [[GitHub Repo](#)]
Tech: Python, Transformers, LangChain, Retrieval-Augmented Generation

CONFERENCE PUBLICATIONS

- Towards Blind and Low-Vision Accessibility of Lightweight VLMs and Custom LLM-Evals:** S. S. Baghel, Y. P. S. Rathore, A. Pradhan. Accepted at the 14th International Joint Conference on Natural Language Processing & the 4th Asian Chapter of ACL (IJCNLP-AACL 2025), Mumbai, India. [[Paper Link](#)]